



HK IMO Training 2023-2024
Problem Set 9
For 9 December 2023



Problem 33.

- (a) Does there exist positive integers a, b, c, d such that $a + b + c = 2020$ and $2^a + 2^b + 2^c = d^2$?
- (b) Does there exist positive integers a, b, c, d such that $a + b + c = 2020$ and $3^a + 3^b + 3^c = d^2$?

Problem 34. There is a circular chessboard with $n \geq 2$ cells arranged in a circle. We put down the numbers 1 to n in the cells in any order such that each number is used exactly once. Initially, we put a chess piece in the cell numbered 1. Every time if the chess piece lies in the cell numbered k , we move the chess piece k steps in the clockwise direction and repeat the same process. Find all possible values of n for which we can write down the numbers in some way such that every cell can be visited at least once after finitely many moves. (Note that the chess piece always visits the cell numbered 1 initially.)

Problem 35. In acute $\triangle ABC$, $AB < AC$. Let D and E be points on the side BC such that D lies between B and E , and $\angle BAD = \angle CAE$. Let F be a point on the side AC . The line BF intersects AD and AE at P and Q respectively. It is known that $DQ \perp AC$ and $EP \perp AB$. Let S be the intersection point of the angle bisectors of $\angle BAC$ and $\angle CBF$. Prove that A, B, D, S, Q are concyclic.

Problem 36.

- (a) Prove that the inequality $(a_1 + a_2 + \cdots + a_n)(b_1 + b_2 + \cdots + b_n) > \frac{2}{9}n^3$ holds for all positive integers n and all pairwise distinct pairs of positive integers (a_i, b_i) (i.e. $(a_i, b_i) \neq (a_j, b_j)$ for $i \neq j$).
- (b) Prove that the constant $\frac{2}{9}$ in part (a) cannot be replaced by any larger constant.